

AI Agent Formulation and Analysis

Selected Agent: Automated Vacuum Cleaner

1. Problem in the Agent's Context

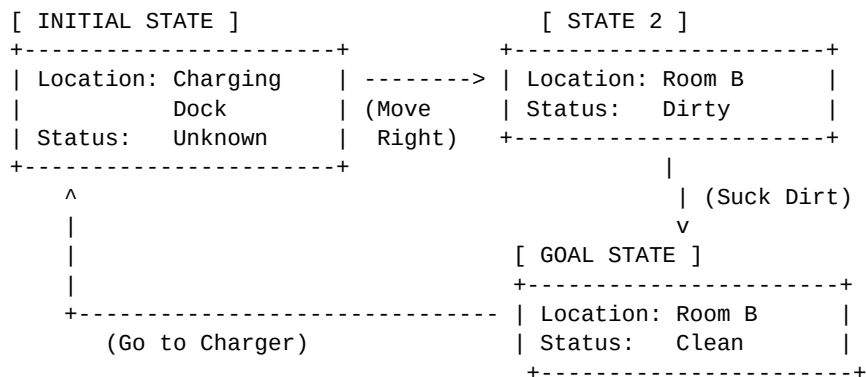
The main problem is to keep a multi-room house floor clean. The agent needs to navigate around randomly placed furniture, avoid falling down stairs, clean up dust or debris, and return to its charging dock before running out of battery.

2. Problem Formulation

Textual Formulation

- **Problem Type:** Multiple-State Problem (Sensorless / Uncertain State) because the agent has uncertain initial knowledge of dirt.
- **States:** The agent's current location (e.g., Room A, Position X) and the status of that location (Clean or Dirty).
- **Initial State:** The vacuum is at the charging dock, and all rooms have an unknown amount of dirt.
- **Actions:** Move Left, Move Right, Move Up, Move Down, Suck Dirt, Go to Charger.
- **Transition Model:** Moving changes the agent's position; sucking dirt changes the room status from Dirty to Clean.
- **Goal Test:** All reachable floor areas are completely clean.
- **Path Cost:** Every move costs 1 point of energy, and sucking dirt costs 2 points. The goal is to maximize cleanliness while using the least energy.

Graphical Formulation



3. PEAS Evaluation

Performance Measure	Percentage of dirt cleaned, total time taken, and battery efficiency.
Environment	Wooden floors, carpets, furniture (chairs/tables), dust, and stairs.
Actuators	Wheels for movement, spinning brushes, and a vacuum suction motor.
Sensors	Bumper sensors (to detect walls), infrared cliff sensors (to avoid stairs), and a dirt detection sensor.

4. Environment Properties

- Partially Observable
- Stochastic
- Sequential
- Dynamic
- Continuous
- Single-Agent